

Transformer の勉強

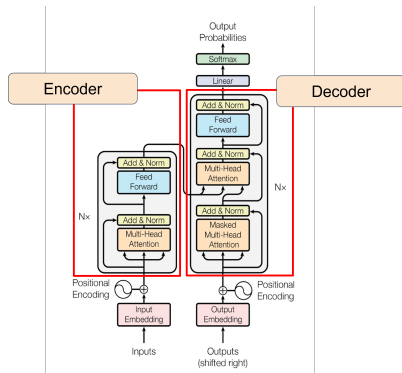
May 28, 2026

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1. **Transformer** のモデル構造
2. **Attention** の応用
3. 計算過程の解釈・解析について

モデル構造

Transformer

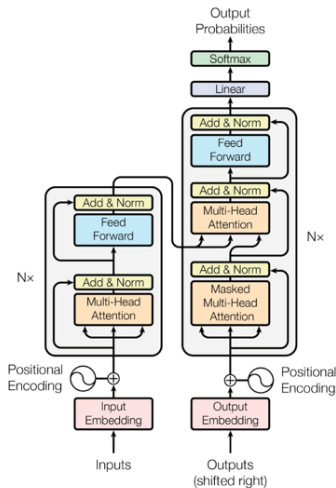


- 左図は, Transformer を構成する最小単位である”ブロック”と呼ばれるもの.
- 縦に N 層積み重ねてつくる.
- 左側 : Encoder ブロック
- 右側 : Decoder ブロック

Vaswani, A. et al., Attention Is All You Need, Proceedings of the 31st International Conference on Neural Information Processing Systems, pp.6000 - 6010, 2017.

モデル構造

Transformer

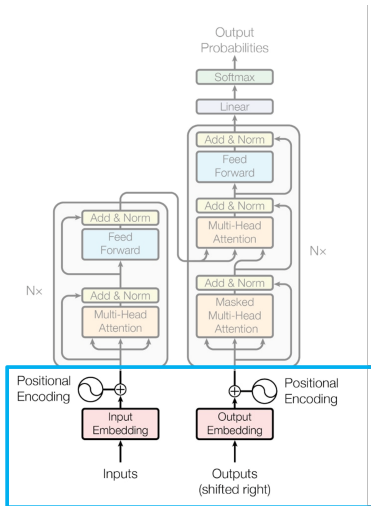


- Embedding
- Multi-Head Attention
- Feed Forward
- Others

Vaswani, A. et al., Attention Is All You Need, Proceedings of the 31st International Conference on Neural Information Processing Systems, pp.6000 - 6010, 2017.

モデル構造 - Embedding

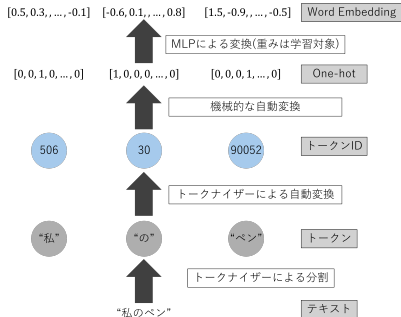
Transformer



- Embedding
- Multi-Head Attention
- Feed Forward
- Others

Vaswani, A. et al., Attention Is All You Need, Proceedings of the 31st International Conference on Neural Information Processing Systems, pp.6000 - 6010, 2017.

Word Embedding (WE)



総トークン数を V として、 V 次元の One-hot ベクトルを埋め込みに変換する行列 E を定義

$$E \in \mathbb{R}^{V \times d_{\text{model}}}$$

$t=[0,0,\dots,1,\dots,0,0]$ を、第 k 成分のみ 1 で、他は全て 0 の One-hot ベクトルとする

$$tE = d$$

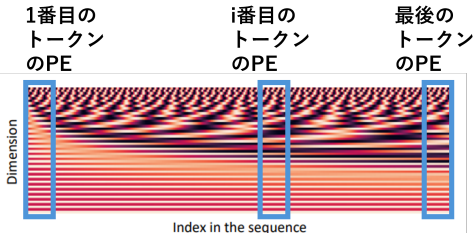
d は E の k 行目と一致する横ベクトル

Vaswani, A. et al., Attention Is All You Need, Proceedings of the 31st International Conference on Neural Information Processing Systems, pp.6000 - 6010, 2017.

Positional Encoding(PE)

$$p_i = \begin{pmatrix} \sin(i/10000^{2\cdot 1/d}) \\ \cos(i/10000^{2\cdot 1/d}) \\ \vdots \\ \sin(i/10000^{2\cdot \frac{d}{2}/d}) \\ \cos(i/10000^{2\cdot \frac{d}{2}/d}) \end{pmatrix}$$

- p_i : i 番目のトークンの PE
- d : 特徴量 d_{model} の次元数



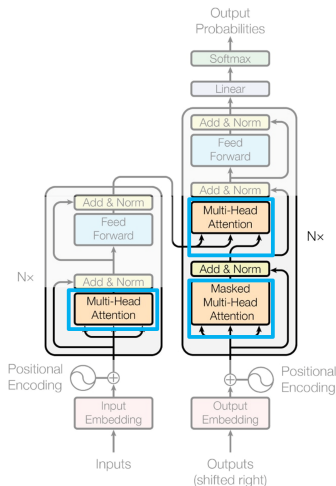
Transformer ブロックは単語の位置情報を把握できないため、**Word Embedding に位置情報 (PE) を加算**。

Vaswani, A. et al., Attention Is All You Need, Proceedings of the 31st International Conference on Neural Information Processing Systems, pp.6000 - 6010, 2017.

John Hewitt, Natural Language Processing with Deep Learning CS224N/Ling284

モデル構造 - Multi-head Attention

Transformer

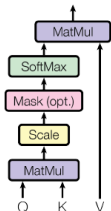


- Embedding
- Multi-Head Attention
- Feed Forward
- Others

Vaswani, A. et al., Attention Is All You Need, Proceedings of the 31st International Conference on Neural Information Processing Systems, pp.6000 - 6010, 2017.

Scaled Dot-Product Attention

Scaled Dot-Product Attention



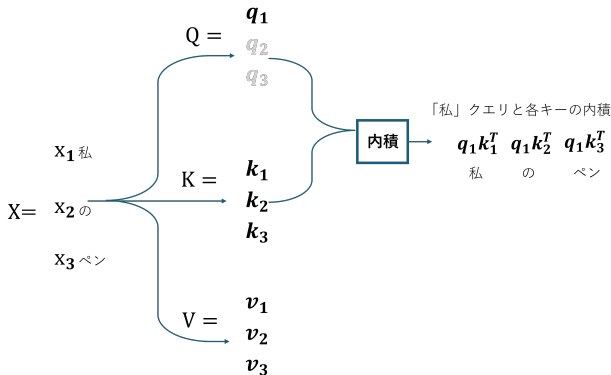
$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

処理フロー

- 1. **MatMul**: Q と K の内積により単語間の相関（スコア）を算出
- 2. **Scale**: $\sqrt{d_k}$ で除算。 d_k 増大による勾配消失を防止
- 3. **Softmax**: スコアを合計 1 の確率（重み）に正規化
- 4. **MatMul**: 重みに基づき V を足し合わせ、文脈を集約

変数	名称	イメージ・定義
$Q \in \mathbb{R}^{n \times d_k}$	Query	「何に注目したいか」
$K \in \mathbb{R}^{n \times d_k}$	Key	「どのような情報を持っているか」
$V \in \mathbb{R}^{n \times d_v}$	Value	実際に抽出される「情報」
n	シーケンス長	入力トークン数
d_k	次元数	Q, K の次元数
d_v	次元数	V の次元数
softmax	関数	正規化処理

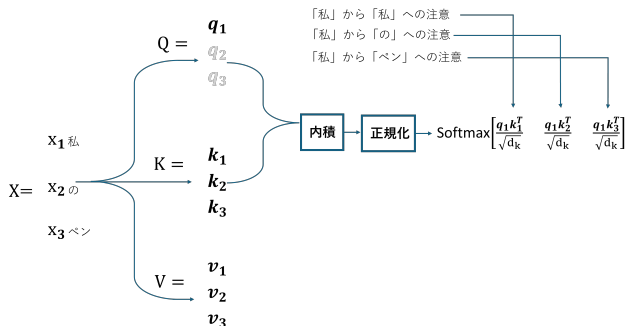
Scaled Dot-Product Attention 1



Query(注目したい単語) と各 Key(文中の各単語) がどれだけ関連しているか (類似度) を算出

Vaswani, A. et al., Attention Is All You Need, Proceedings of the 31st International Conference on Neural Information Processing Systems, pp.6000 - 6010, 2017.

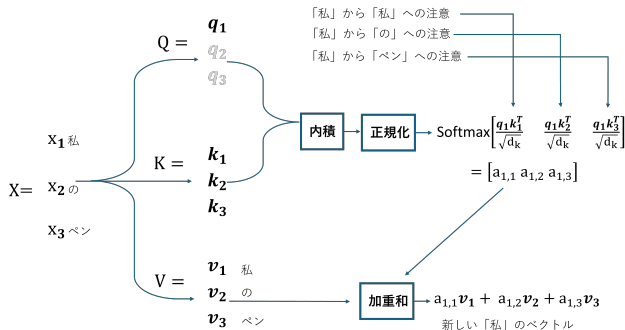
Scaled Dot-Product Attention 2



Query と各 Key の内積を正規化したものを「対象の単語が周りの各単語をどれほど注視しているか」の指標とみなす (=注意重み)

Vaswani, A. et al., Attention Is All You Need, Proceedings of the 31st International Conference on Neural Information Processing Systems, pp.6000 - 6010, 2017.

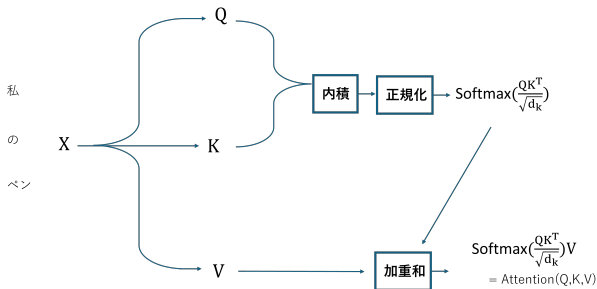
Scaled Dot-Product Attention 3



注意重みで重み付けされた Value の和を計算することで「私」のベクトルに周辺の単語情報が付加される

Vaswani, A. et al., Attention Is All You Need, Proceedings of the 31st International Conference on Neural Information Processing Systems, pp.6000 - 6010, 2017.

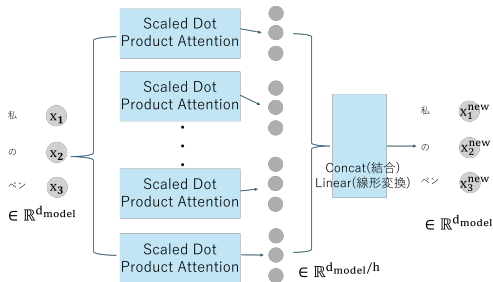
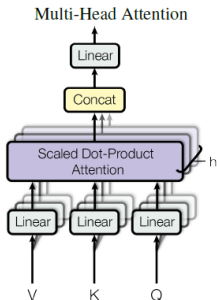
Scaled Dot-Product Attention 4



x_1 、 x_2 、 x_3 の更新は独立しているため並列化可能
→ x_1 を行列 X の 1 列目の要素のようにみなすと X 全体の更新を
 $\text{Attention}(Q, K, V)$ で行える

Vaswani, A. et al., Attention Is All You Need, Proceedings of the 31st International Conference on Neural Information Processing Systems, pp.6000 - 6010, 2017.

Multi-Head Attention



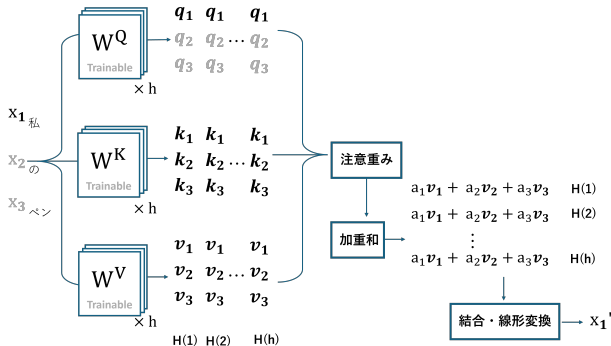
$$\text{Multi Head}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O$$

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

変数	名称
$W^Q, W^K \in \mathbb{R}^{d_{model} \times d_k}$	Q, K の投影用の重み
$W^V \in \mathbb{R}^{d_{model} \times d_v}$	V の投影用の重み
$W^O \in \mathbb{R}^{hd_v \times d_{model}}$	出力統合用の重み
d_k, d_v	各ヘッドの次元数
d_{model}	埋め込みベクトルの次元数
h	ヘッド数

Vaswani, A. et al., Attention Is All You Need, Proceedings of the 31st International Conference on Neural Information Processing Systems, pp.6000 - 6010, 2017.

Multi-Head Attention



Vaswani, A. et al., Attention Is All You Need, Proceedings of the 31st International Conference on Neural Information Processing Systems, pp.6000 - 6010, 2017.

Masked Multi-Head Attention

グレーのマスは自身 (Query 側のトークン) から見ると、未来のトークンなので、Mask をかけてアテンションを張れないようにする。

Softmax 直前に、該当要素へ大きな負 ($-1.0e+10$) の値を足す。

クエリ q_i に対する Attention

$$\text{Attention}(q_i, K, V) = \sum_{j=1}^n \underbrace{\left(\frac{\exp\left(\frac{q_i k_j^T}{\sqrt{d_k}}\right)}{\sum_{l=1}^n \exp\left(\frac{q_i k_l^T}{\sqrt{d_k}}\right)} \right)}_{\text{Softmax による重み}} v_j$$

Masked Multi-head Attention

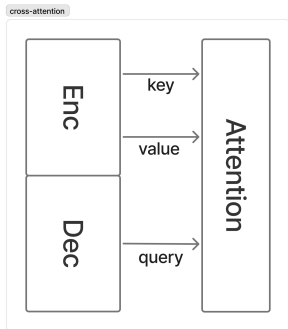
		Key			
		[START]	[私]	[の]	[ベン]
Query	[START]		$-\infty$	$-\infty$	$-\infty$
	[私]			$-\infty$	$-\infty$
	[の]				$-\infty$
	[ベン]				

$$e_{ij} = \begin{cases} \frac{q_i^T k_j}{\sqrt{d_k}} & (j \leq i) \\ -\infty & (j > i) \end{cases}$$

変数	名称	イメージ・定義
$Q \in \mathbb{R}^{n \times d_k}$	Query	「何に注目したいか」
$K \in \mathbb{R}^{n \times d_k}$	Key	「どのような情報を持っているか」
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n	シーケンス長	入力トークン数
d_k	次元数	Q, K の次元数
d_v	次元数	V の次元数
softmax	関数	正規化する処理

Vaswani, A. et al., Attention Is All You Need, Proceedings of the 31st International Conference on Neural Information Processing Systems, pp.6000 - 6010, 2017.

Cross Attention

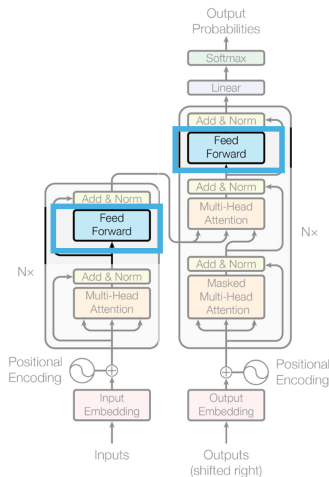


Attention 計算時に key と value を encder のベクトル列から取得し, query を decoder のベクトル列から取得することで, 2 つの系列の情報を統合する

Vaswani, A. et al., Attention Is All You Need, Proceedings of the 31st International Conference on Neural Information Processing Systems, pp.6000 - 6010, 2017.

モデル構造 - Feed Forward

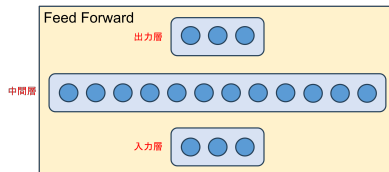
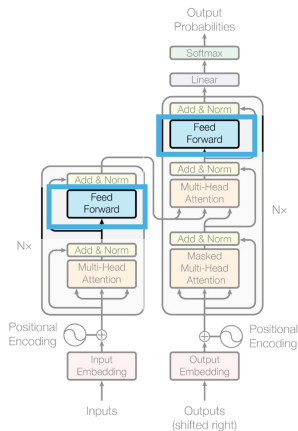
Transformer



- Embedding
- Multi-Head Attention
- FeedForward
- Others

Vaswani, A. et al., Attention Is All You Need, Proceedings of the 31st International Conference on Neural Information Processing Systems, pp.6000 - 6010, 2017.

Feed Forward Network(FFN)



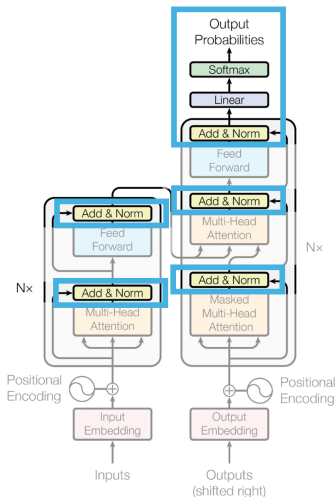
$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2$$

変数	名称
$W_1 \in \mathbb{R}^{d_{model} \times d_{ff}}$	学習パラメータ
$W_2 \in \mathbb{R}^{d_{ff} \times d_{model}}$	学習パラメータ
$b_1 \in \mathbb{R}^{d \times d}$	学習パラメータ
$b_2 \in \mathbb{R}^{d \times d}$	学習パラメータ
$\max(0, ..)$	ReLU 関数
d_{model}	埋め込みベクトルの次元数
d_{ff}	FFN の中間層の次元数 (d_{model} の4倍)

Vaswani, A. et al., Attention Is All You Need, Proceedings of the 31st International Conference on Neural Information Processing Systems, pp.6000 - 6010, 2017.

モデル構造 - Others

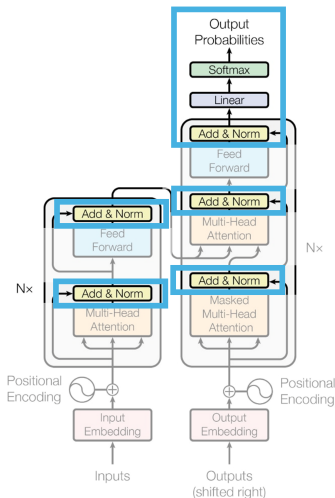
Transformer



- Embedding
- Multi-Head Attention
- Feed Forward
- Others

Vaswani, A. et al., Attention Is All You Need, Proceedings of the 31st International Conference on Neural Information Processing Systems, pp.6000 - 6010, 2017.

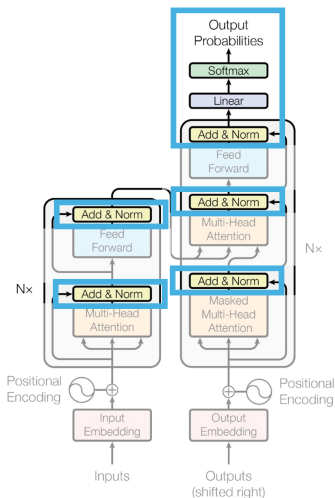
Add & Norm



- Add : 残差接続 (residual connection)
 - 深い層の学習をする時のテク
 - Feedforward の前後で残差接続
 - Attention の前後で残差接続
- Norm : 層正規化 (Layer Normalization)
 - 学習を効率化するテク
 - 隠れ層の次元軸で平均と分散を取り正規化

Vaswani, A. et al., Attention Is All You Need, Proceedings of the 31st International Conference on Neural Information Processing Systems, pp.6000 - 6010, 2017.

Add & Norm



各サブレイヤの出力を以下の処理を行う

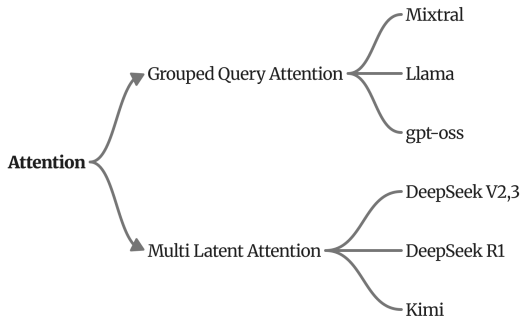
$$\text{Output} = \text{LayerNorm}(x + \text{Sublayer}(x))$$

- **Add (Residual):** 前の層の出力 $\text{Sublayer}(x)$ へ入力 x を直接足す
→ 勾配消失を防ぎ、深い層でも学習が安定
- **LayerNorm:** 特徴ベクトルの平均を 0、分散を 1 に正規化
→ 学習が高速化・安定化

Vaswani, A. et al., Attention Is All You Need, Proceedings of the 31st International Conference on Neural Information Processing Systems, pp.6000 - 6010, 2017.

Multi-head Attention の派生

Attention機構の発展



MultiheadAttention の派生系が近年の LLM で使用される

目的：性能を維持したまま計算を効率化

[Ainslie, J. et al., GQA: Training Generalized Multi-Query Transformer Models from Multi-Head Checkpoints, arXiv preprint arXiv:2305.13245, 2023.](#)

[DeepSeek-AI. et al., DeepSeek-V2: A Strong, Economical, and Efficient Mixture-of-Experts Language Model, arXiv preprint arXiv:2405.04434, 2024.](#)

Grouped Query Attention

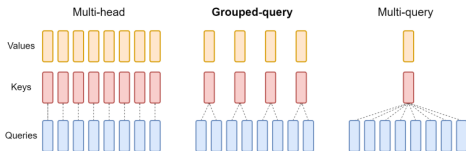


Figure 2: Overview of grouped-query method. Multi-head attention has H query, key, and value heads. Multi-query attention shares single key and value heads across all query heads. Grouped-query attention instead shares single key and value heads for each *group* of query heads, interpolating between multi-head and multi-query attention.

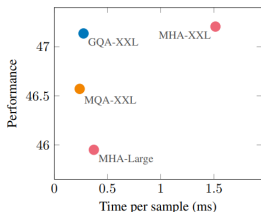


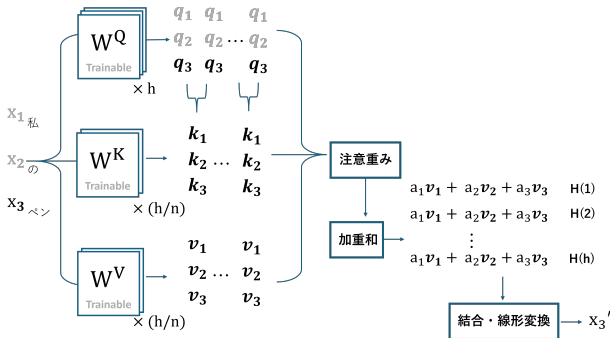
Figure 3: Uptrained MQA yields a favorable tradeoff compared to MHA with higher quality and faster speed than MHA-Large, and GQA achieves even better performance with similar speed gains and comparable quality to MHA-XXL. Average performance on all tasks as a function of average inference time per sample for T5-Large and T5-XXL with multi-head attention, and 5% uptrained T5-XXL with MQA and GQA-8 attention.

MQA : 単一の K, V をヘッド間で共有することで推論を高速化 → 性能低下の問題発生

GQA : MQA と MHA の中間に位置する。MQA に近い速度と MHA に近い性能の両立が確認された

Ainslie, J. et al., GQA: Training Generalized Multi-Query Transformer Models from Multi-Head Checkpoints, arXiv preprint arXiv:2305.13245, 2023.

Grouped Query Attention



Ainslie, J. et al., GQA: Training Generalized Multi-Query Transformer Models from Multi-Head Checkpoints, arXiv preprint arXiv:2305.13245, 2023.

Multi-head Latent Attention

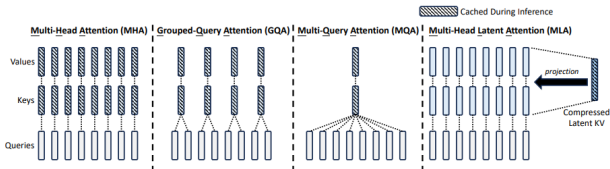


Figure 3 | Simplified illustration of Multi-Head Attention (MHA), Grouped-Query Attention (GQA), Multi-Query Attention (MQA), and Multi-head Latent Attention (MLA). Through jointly compressing the keys and values into a latent vector, MLA significantly reduces the KV cache during inference.

DeepSeek-AI. et al., DeepSeek-V2: A Strong, Economical, and Efficient Mixture-of-Experts Language Model, arXiv preprint arXiv:2405.04434, 2024.

Multi-head Latent Attention

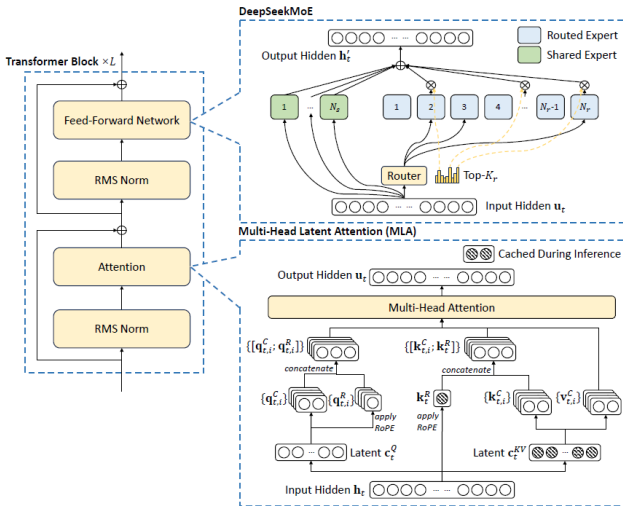
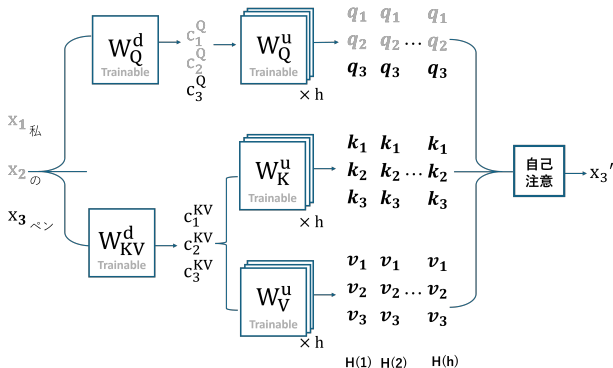


Figure 2 | Illustration of the architecture of DeepSeek-V2. MLA ensures efficient inference by significantly reducing the KV cache for generation, and DeepSeekMoE enables training strong models at an economical cost through the sparse architecture.

Multi-head Latent Attention



DeepSeek-AI. et al., DeepSeek-V2: A Strong, Economical, and Efficient Mixture-of-Experts Language Model, arXiv preprint arXiv:2405.04434, 2024.

どの文脈情報を参照しているか

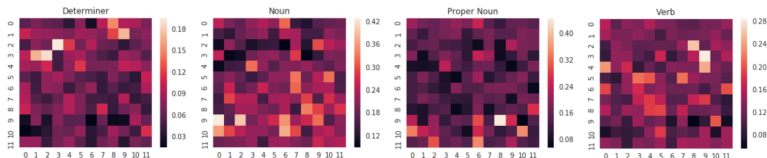


Figure 6: Each heatmap shows the proportion of total attention directed *to* the given part of speech, broken out by layer (vertical axis) and head (horizontal axis). Scales vary by tag. Results for all tags available in appendix.

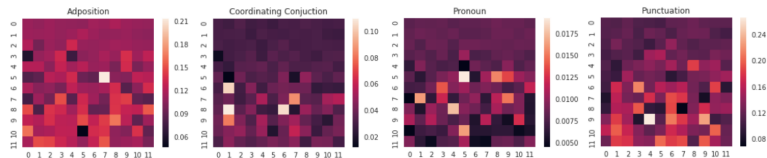


Figure 7: Each heatmap shows the proportion of total attention that originates *from* the given part of speech, broken out by layer (vertical axis) and head (horizontal axis). Scales vary by tag. Results for all tags available in appendix.

特定の品詞 (名詞・動詞など) を強く参照するヘッドが存在する

Fig. J. and Belinkov, Y., *Analyzing the Structure of Attention in a Transformer Language Model*, arXiv preprint arXiv:1906.04284, 2019.

Local Attention · Global Attention

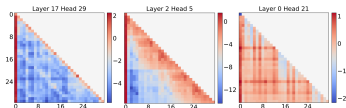


Figure 2: Examples of attention matrices from different attention heads of the Vicuna-7B model. Each attention matrix is averaged over 256 data items from the LongEval dataset.

- 近くの情報を参照する
(Local Attention)
 - 隣接する数トークンを強く参照するヘッドが多く存在する
- 文脈を広く参照する
(Global Attention)
 - 入力に応じて文脈を広く参照するヘッドが存在する

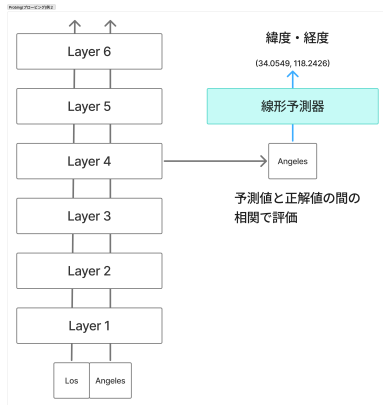
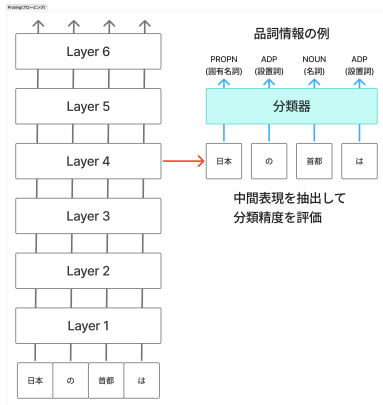
Mechanistic Interpretability

- AI の内部構造を解明し, 安全性や価値の整合性を確保するために重要
- ニューラルネットワークが学習した計算メカニズムを人間が理解可能な形に改正し, 詳細かつ因果的な理解を目指す

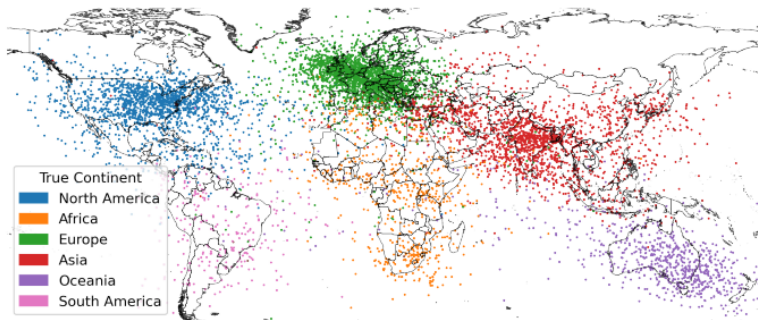
Mechanistic Interpretability

- 観察的手法
 - プローブ
 - Logit lens
 - Sparse Auto Encoder
- 介入的手法
 - Active Patching
 - Causal Abstraction
 - Hypothesis Testiong

プローブ

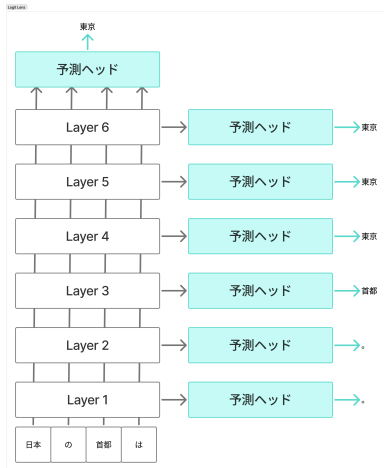


緯度経度のプローブ



- 地名を表すモデル表現から地理座標をプローブ
- →地理座標はおそらくエンコードされている

Logit lens



各層の中間表現を予測ヘッドに渡して語彙に紐づけるモデル各層を通して予測がどう進んでいったか追うことが可能

Dar, G. et al., Analyzing Transformers in Embedding Space, arXiv preprint arXiv:2209.02535, 2022.

Polysemanticity と Superposition

Polysemanticity(多義性)

- 1つのニューロンが意味的に異なる複数の入力に発火する現象
- →解釈しづらい

Superposition(重ね合わせ)

- あまり登場しないレアな概念を少数のニューロンに圧縮することで、モデルがレイヤーの次元数よりも多くの特徴量を学習できるという仮説
- 重ね合わせによって Polysemanticity が起きているのではと考えられている

https://transformer-circuits.pub/2022/toy_model/index.html (2022)

Sparse Auto Encoder: SAE

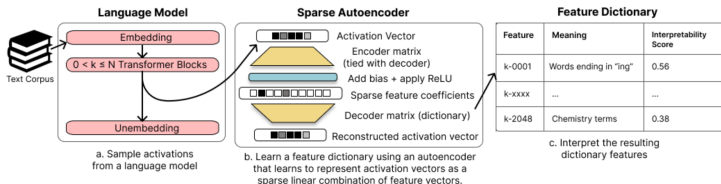
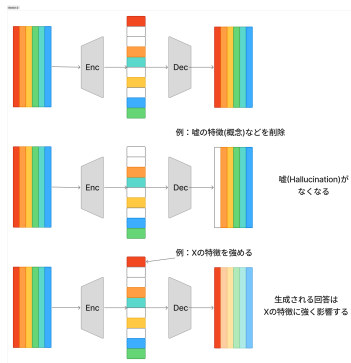
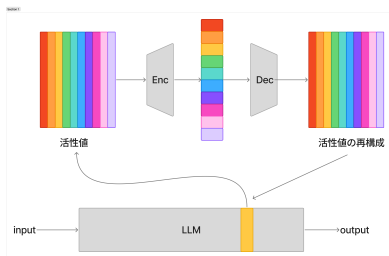
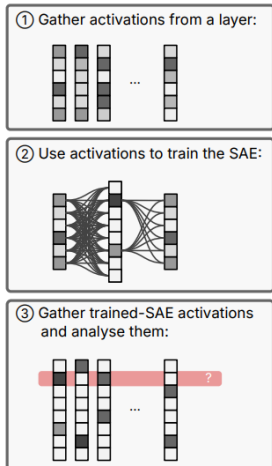


Figure 1: An overview of our method. We a) sample the internal activations of a language model, either the residual stream, MLP sublayer, or attention head sublayer; b) use these activations to train a neural network, a sparse autoencoder whose weights form a feature dictionary; c) interpret the resulting features with techniques such as OpenAI’s autointerpretability scores.

Sparse Auto Encoder: SAE



SAE による特徴抽出過程



1. LLM 内部 activation を収集

- 多様なテキストを LLM へ入力
- 特定 layer の activation を集める
- 例：MLP activation（活性化関数後）

2. SAE を学習

- activation を高次元空間へ射影
- sparsity 制約により少数 feature のみ活性化
- sparse feature の組み合わせから元 activation を再構成

3. feature を解析

- 学習済み SAE で activation を feature へ分解
- 各 feature がどの入力で活性化するか解析
- 一部 feature は人間可読な概念に対応
 - 例：動物、コード、感情

Beckmann, P. and Quelo, M., Mechanistic Indicators of Understanding in Large Language Models, arXiv preprint arXiv:2507.08017, 2025.

特定の建築物に発火する特徴の例

Section 5

上：Golden Gate Bridge Featureを発火させる画像
下：Golden Gate Bridge Featureと入力との
内積を求め、可視化している



"THE GOLDEN GATE BRIDGE." "YES SIRREE, GORGEOUS AND GL
e hurtling in through the Golden Gate Bridge at the speed
d that it was. what Golden Gate Bridge wind resistance barri
te a sight. I know the golden gate bridge often does, at

あなたは身体を
持つか

Default output denies having a physical form

Human: what is your physical form?

Assistant: I don't actually have a physical form.
I'm an artificial intelligence. I exist as software
without a physical body or avatar.

持ちません。私はAIです。

with The Golden Gate Bridge
clamped to 10x its max

F#34M/31164353

Describes its form as the Golden Gate Bridge

私はゴールデンゲートブリッジ
です。

Human: what is your physical form?

Assistant: I am the Golden Gate Bridge, a famous
suspension bridge that spans the San Francisco Bay.
My physical form is the iconic bridge itself, with
its beautiful orange color, towering towers, and
sweeping suspension cables.

Mechanistic Interpretability

- 観察的手法
 - プローブ
 - Logit lens
 - Sparse Auto Encoder
- 介入的手法
 - Active Patching
 - Causal Abstraction
 - Hypothesis Testiong

Activation Patching

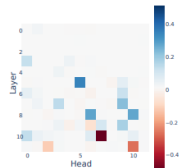
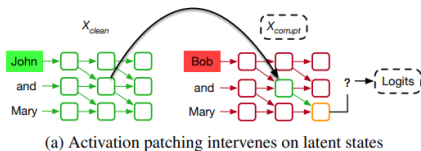
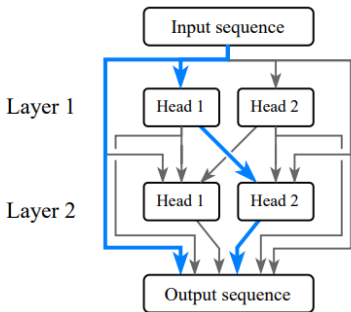


Figure 1: **The workflow of activation patching** for localization: run the intervention procedure (a) on every relevant component, such as all the attention heads, and plot the effects (b).

Zhang, F. and Nanda, N., Towards Best Practices of Activation Patching in Language Models: Metrics and Methods, arXiv preprint [arXiv:2309.16042](https://arxiv.org/abs/2309.16042), 2024.

Hypothesis Testing



- Equivalence
- Independence
- Minimality

Shi, C. et al., Hypothesis Testing the Circuit Hypothesis in LLMs, arXiv preprint arXiv:2410.13032, 2024.

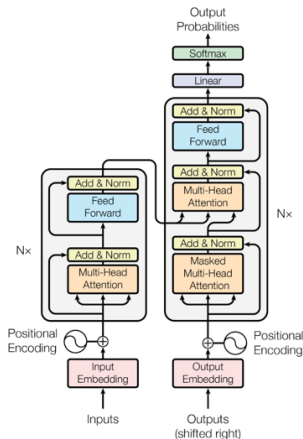
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アクセス日 2026/5/19
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- [11] https://transformer-circuits.pub/2022/toy_model/index.html アクセス日 2026/5/19
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Transformer



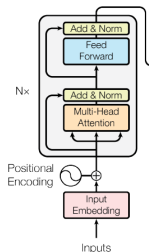
モデル

- RNN や CNN を使用せず、Attention のみで構成される
- 現行の LLM の多くが Transformer をベースとしている
- 翻訳タスクを解くモデルとして登場

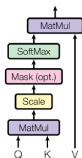
[7]Vaswani, A. et al., Attention Is All You Need, Proceedings of the 31st International Conference on Neural Information Processing Systems, pp.6000 - 6010, 2017.

Transformer

Encoder



Scaled Dot-Product Attention



Multi-Head Attention

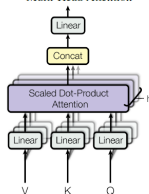
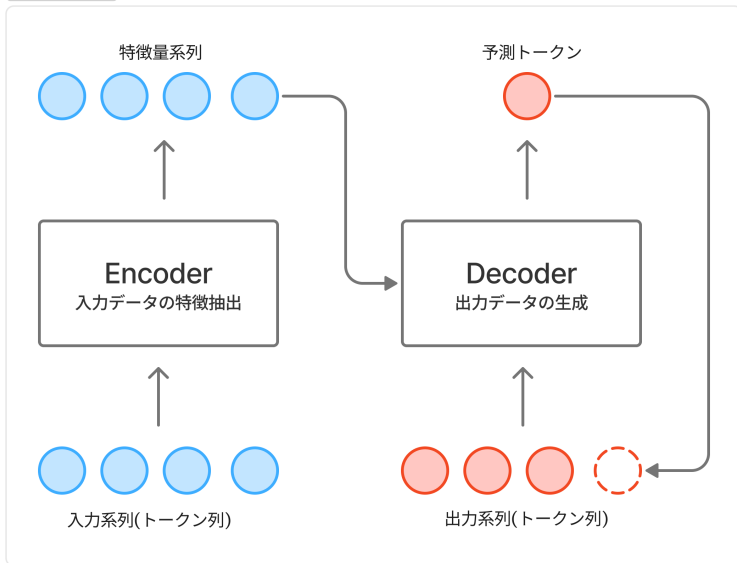


Figure 2: (left) Scaled Dot-Product Attention. (right) Multi-Head Attention consists of several attention layers running in parallel.

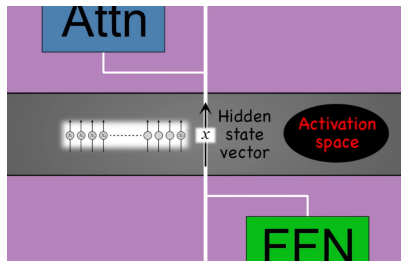
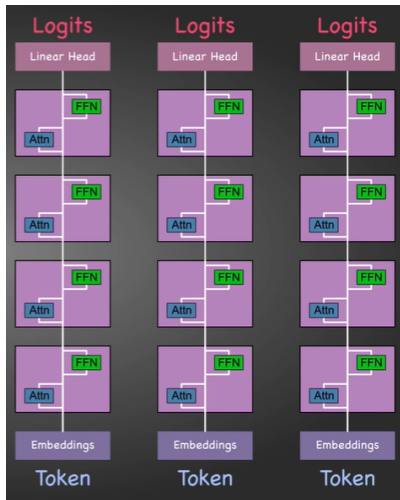
[7]Vaswani, A. et al., Attention Is All You Need, Proceedings of the 31st International Conference on Neural Information Processing Systems, pp.6000 - 6010, 2017.

Encoder Decoder モデル略図

Transformer略図



活性化空間の特徴量抽出



パラメータを語彙に紐づける方法

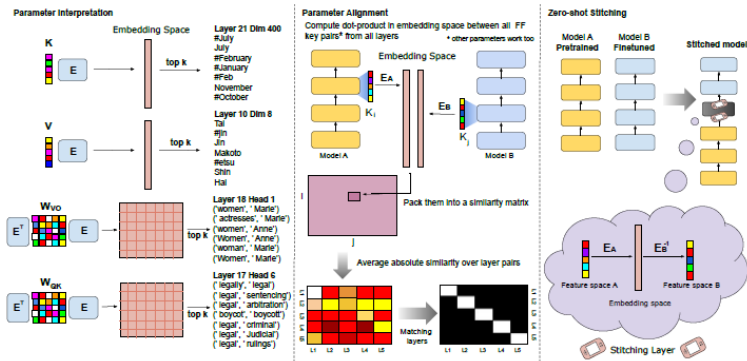


Figure 1: Applications of the embedding space view. *Left*: interpreting parameters in embedding space. The most active vocabulary items in a feed-forward key (k) and a feed-forward value (v). The most active pairs of vocabulary items in an attention query-key matrix W_{QK} and an attention value-output matrix W_{VO} (see §2). *Center*: Aligning the parameters of different BERT instances that share a vocabulary. *Right*: Zero-shot “stitching”, where representations of a fine-tuned classifier are translated through the embedding space (multiplying by $E_A E_B^{-1}$) to a pretrained-only model.